

This project aims to develop a lightweight, real-time network security visualization tool inspired by SIEM functionalities using Observable. The tool will provide an interactive interface for visualizing network traffic anomalies, detecting potential denial-of-service (DoS) attacks, and monitoring suspicious activity in a controlled test environment.

The intended users for this system are cybersecurity professionals, SOC analysts, penetration testers, and security researchers who require an easy-to-deploy and interactive visualization tool for monitoring network security.

Users are expected to have:

Intermediate technical knowledge of networking

Basic familiarity with visualization tools and security data analysis.

Experience with SIEMs and IDS/IPS systems is helpful but not required.

We will generate our own dataset through a controlled cybersecurity test environment:

Red Team (Attacker): A Kali Linux machine will execute nmap scans, DoS attacks (SYN flood, ICMP flood)

Blue Team (Defender): A Linux-based machine will run Wireshark/tshark to capture real-time network traffic logs.

Inline IDS (Optional): Firewall rules or simple packet filtering scripts will be used to detect attack patterns.

Network Environment: A self-contained private LAN setup using a mobile hotspot or a VLAN configuration to isolate attack traffic from external networks.

To facilitate real-time visualization, we will:

Stream Wireshark logs via WebSockets into Observable, enabling real-time network monitoring.

Parse attack-specific features (e.g., packet spikes, SYN flood detection) to visualize network anomalies dynamically.

The proposed SIEM will feature multiple interactive views to explore network activity:

Real-Time Traffic Monitoring Dashboard

Live packet count visualization to detect abnormal traffic spikes.

Brushing & filtering features to zoom in on specific time windows.

Interactive Packet Drill-Down

Clickable data points that reveal detailed packet-level insights (source IP, destination IP, protocol, request type).

Blue Team (Defender) - VLAN Setup & Network Visualization

1. Request Rules of Engagement (RoE)

- Email the professor with:
 - **Objective of the test**
 - **What tools you will use**
 - **Safety precautions**
 - If **denied**: Use **mobile hotspot + VLAN isolation** instead of SFU Wi-Fi.
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2. Prepare Kali Setup

Backup Your Kali Install (Optional)

```
sudo dd if=/dev/sda of=/mnt/backup/kali_backup.img bs=4M
```

Install Necessary Tools

```
sudo apt update && sudo apt install wireshark tshark apache2 -y
```

3. Test Observable-to-Kali Data Flow

Check WebSockets Connection

- Run a Python WebSocket server:

```
pip install websockets
```

```
python3 -m websockets ws://localhost:8080
```

- If error, check firewall:

```
sudo ufw allow 8080
```

4. Purchase & Setup Network Hardware

- Buy:
 - **USB-to-Ethernet adapter**
 - **Ethernet cable**
- **Enable Network Interface:**

```
sudo ifconfig eth0 up
```

```
sudo dhclient eth0
```

- Check connection:

```
ping -c 4 8.8.8.8
```

5. Configure VLAN with WAN Access

Check Available Interfaces

```
ip link show
```

Create VLAN

```
sudo ip link add link eth0 name eth0.10 type vlan id 10
```

Assign IP to VLAN

```
sudo ip addr add 192.168.1.100/24 dev eth0.10
```

Enable Interface

```
sudo ip link set eth0.10 up
```

Test Internet on VLAN

```
ping -I eth0.10 8.8.8.8
```

6. Test VLAN with Armitage (Red Team)

Step 1: Ensure Isolation

```
sudo tcpdump -i eth0.10
```

Step 2: Simulate Attack & Monitor

- Run **Armitage scan** from Red Team:

```
nmap -A -T4 -v -Pn 192.168.1.0/24
```

- Check if Blue Team **logs packets**:

```
tshark -i eth0.10
```

7. Backend Attack Simulation (Vertical Slice)

- Run **30s Normal Traffic**

```
ping -c 30 8.8.8.8
```

- Run **1s ICMP Flood**

```
hping3 --flood --icmp -d 120 -p 80 192.168.1.200
```

- Run **1s SYN Flood**

```
hping3 -S --flood -V -p 80 192.168.1.200
```

- Capture Data:

```
tshark -i eth0.10 -w attack_test.pcap
```

8. Frontend Visualization & Testing

- Check if **data is displayed** in Observable.
- Run **frontend GUI** in browser:

```
firefox http://localhost:8080
```

- Test with normal traffic:

```
curl -X GET http://localhost:8080
```

9. Full Integration Test

- Meet with **teammate**.
- **Run full test** (Red Team attacks, Blue Team monitors).
- Verify **live visualization updates**.

10. Live Demo in Class

- Start **full test setup**:
 - **Armitage Attack**
 - **Wireshark Capture**
 - **Real-Time Visualization**
- Explain findings.

Red Team (Attacker) - Kali Linux Setup on Windows with VirtualBox

1. Install VirtualBox and Kali Linux

Download Required Software

- **VirtualBox:** [Download Here](#)
- **Kali Linux ISO:** [Download Here](#) (you can just grab this with the virtual box option)

Create a New Virtual Machine in VirtualBox (optional)

1. Open **VirtualBox** and click **New**.
2. **Set Name & Type:**
 - Name: Kali-Red
 - Type: Linux
 - Version: Debian (64-bit)
3. **Allocate Memory:** Set RAM to **4096MB (4GB)**.
4. **Create Virtual Hard Disk:**
 - Select **Create a virtual hard disk now**.
 - Choose **VDI (VirtualBox Disk Image)**.
 - Storage: **Dynamically allocated**.
 - Size: **40GB**.

5. Click **Create**.
 6. Go to **Settings** → **Storage**, select the **Empty disk**, and attach the Kali Linux ISO.
 7. Start the VM and follow **on-screen instructions** to install Kali Linux.
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2. Install Armitage for GUI-Based Pentesting

Update System Packages

```
sudo apt update && sudo apt upgrade -y
```

Install Armitage and Metasploit

```
sudo apt install armitage -y
```

```
sudo msfdb init
```

Start Metasploit Database

```
sudo systemctl start postgresql
```

```
sudo msfconsole
```

Start Armitage

```
armitage
```

3. Learn Basic Scanning & Attacks

Nmap Scan (Network Reconnaissance)

```
nmap -A -T4 -v [TARGET_IP]
```

- -A: Aggressive mode (OS detection, version detection)
- -T4: Faster execution
- -v: Verbose mode

ICMP Flood Attack (Simulated DDoS)

```
hping3 --flood --icmp -d 120 -p 80 [TARGET_IP]
```

SYN Flood (DoS Attack)

```
hping3 -S --flood -V -p 80 [TARGET_IP]
```

4. Troubleshooting Commands

Check Network Connectivity

`ifconfig`

`ping -c 4 google.com`

Restart VirtualBox Network

`sudo systemctl restart networking`

Check Metasploit Database Status

`sudo systemctl status postgresql`

5. Final Steps

1. Ensure **Kali VM is running smoothly** inside VirtualBox.
2. **Test Armitage** to confirm it can launch properly.
3. Prepare for **hands-on attack simulations** under controlled conditions.